Fiscal Year:	FY 2026	**Task Last Updated:**	FY 12/22/2025
PI Name:	Williams, Michael A. M.D.		
Project Title:	Zero G and ICP: Invasive and Noninvasive ICP Monitoring of Astronauts on the ISS		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-ILSRA. International Life Sciences Research Announcement
Start Date:	02/05/2016	**End Date:**	01/31/2031
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 01/31/2031 per V. Lehman/NASA-JSC (Ed., 12/20/24). Title reverted to “Zero G and ICP: Invasive and Noninvasive ICP Monitoring of Astronauts on the ISS” per NSSC documentation (Ed., 12/20/24). NOTE: End date changed to 1/31/2025 per NSSC information (Ed., 11/7/19) NOTE: Title change to "Invasive and Noninvasive ICP Monitoring and SANS Biomarker Identification" per PI (Ed., 12/20/18) NOTE: Title change to "Invasive and Noninvasive ICP Monitoring and VIIP Biomarker Identification". Previous title was "'Zero G and ICP: Invasive and Noninvasive ICP Monitoring of Astronauts on the ISS" (Ed., 8/22/16).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Eklund, Anders Ph.D. (Umea University, Sweden) Malm, Jan M.D., Ph.D. (Umea University, Sweden) Bershad, Eric M.D. (Baylor College of Medicine) Furman, David (The Buck Institute for Research on Aging)		
Grant/Contract No.:	NNX16AE78G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Per the Principal Investigator (PI), protocol modifications submitted in May 2025 and November 2025 were approved by the NASA Institutional Review Board (IRB). See the FY2026 report for specific changes to protocol (Ed., 1/22/26). Project combined with Zanello's "Multimodal Modeling towards Noninvasive Assessment of Intracranial Pressure in Weightlessness and Biomarker Identification of Predisposition to VIIP Syndrome" (Ed., 8/22/16). The primary purpose of this study is to determine whether abnormal intracranial pressure (ICP) is related to the development of Spaceflight Associated Neuro-Ocular Syndrome (SANS). This study will evaluate ICP in astronauts after spaceflight using a lumbar puncture (LP), also known as a spinal tap, which is a reliable, commonly used method for measuring ICP. Although methods to estimate ICP without a spinal tap (called noninvasive ICP) exist, they are not yet accurate enough to be reliable, and they are unable to assess important fluctuations in the ICP. Blood, urine, and cerebrospinal fluid (CSF) samples are analyzed for candidate biomarkers that correspond with changes in ICP, and to investigate the molecular basis and genetic predisposition for developing SANS. Research has shown that the presence of certain social, psychological, environmental, and biological factors has been linked to the promotion of a state of low-grade, non-infectious, i.e., sterile, systemic chronic inflammation (SCI) that is characterized by the activation of immune components that are often distinct from those engaged during an acute immune response. We hypothesize that a state of SCI can be seen in astronauts who develop SANS and that this will co-exist with a state of lowered immunity. To assess changes in immune and molecular states that occur in astronauts during spaceflight, we will compare their results to data derived from a large survey of immunity and aging, the Stanford 1,000 Immunomes Project (Stanford 1KIP). We will measure the presence of auto-antibodies and viral antibodies as well. We will screen the biomarkers associated with SANS using in silico methods [STITCH, Search Tool for Interactions of Chemicals (stitch.embl.de) and the NIH Library of Integrated Network-Based Cellular Signatures initiative, LINCS L1000 (NIH LINCS Program (lincsproject.org))] to identify compounds that may have the potential to prevent development of SANS. In summary, the purpose of this study is to determine if abnormal ICP is associated with the development of SANS, and to determine if analysis of blood, urine, or CSF can identify which subjects will develop SANS. Predictive models will be built using data collected from ICP measures and the biomarkers to determine the probability a crewmember may develop SANS. OBJECTIVES 1) Determine whether ICP and related indicators of craniospinal compliance after return to Earth after long duration spaceflight differ from Pre-Flight baseline values and whether immediate post-flight ICP differs from ICP measured 6 months after return to Earth using NASA-approved direct ICP methods (lumbar puncture). 2) Determine the correlation of post-flight ICP changes to standard clinical indicators of SANS. 3) Use novel methods of mRNA and microRNA from RNA sequencing to identify candidate urinary, CSF and plasma biomarkers that align with SANS-related physiologic variables. 4) Assess changes in immunological and molecular trajectories associated with systemic chronic inflammation (SCI) in astronauts during spaceflight. We will project each crew member’s profile on the topological representation of the multidimensional immune network from the Stanford 1KIP to evaluate proximity or deviations from health pre-flight, in-flight, and post-flight. 5) Identify compounds that could ameliorate or reverse the immunologic or metabolic mechanisms identified by the biomarkers associated with SANS. 6) Determine whether there are detectable differences between folate intermediary levels in tissue specific derived bio-fluids and systemic fluids, and determine whether there is a depletion of folate metabolites in the CSF that may indicate impairment of the blood brain barrier in response to spaceflight SIGNIFICANCE: This application addresses the following Integrated Research Plan (IRP) Gaps: SANS01 - etiological mechanisms and contributing risk factors, SANS03 - validated and minimally obtrusive diagnostic tools to measure and monitor changes in ICP, and SANS12 - validation of ground-based analogs. This approach has the potential to determine if Spaceflight Associated Neuro-Ocular Syndrome (SANS) is related to alterations of ICP and, if so, to initiate research into prevention or treatment countermeasures and, if not, to focus research efforts on other potential etiologies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gaining knowledge about the adaptive changes to the space environment will help to further characterize human health and performance risks and the development of countermeasures. Identification of biomarkers associated with recovery from long duration spaceflight, especially if they correlate with the development and resolution of clinical findings of SANS could result in clinical biomarkers to follow the development or recovery from treatment of intracranial hypertension, hydrocephalus, or other disorders of CSF circulation in patients on earth.
	The following key tasks were accomplished in 2025: 1) Protocol modifications submitted in May 2025 and November 2025 were approved by the NASA Institutional Review Board (IRB). 2) A Technology Readiness Review was performed in May 2025 and approved in May 2025. 3) At the request of the Chief Scientist’s Office and an astronaut who wished to have both pre-flight and post-flight lumbar puncture protocols, we modified the protocol to reinstate the original pre /post-flight protocol, and received approval from the IRB. This is now known as Arm 3. The modifications in in November were primarily to reflect changes to the protocol secondary to reduced activities on the International Space Station (ISS). For example, all specimen collections aboard the ISS were removed from the

Task Progress:	protocol. The modifications also reflect the decision to close the protocol to further enrollment. Arm 1 is the existing protocol and was not changed. 4) For Arm 3, an astronaut consented to the protocol and the pre-flight protocol was performed in 2025. 5) One astronaut has consented to participate in Arm 1 of the protocol and is currently aboard the ISS. The post-flight lumbar puncture protocol is expected in 2026. 6) The NASA Human Research Program (HRP) capped the enrollment at 3, including the single astronaut from 2024 and the 2 currently enrolled astronauts described above. No more study enrollment will occur. The study is expected to conclude in September 2027.
Bibliography Type:	Description: (Last Updated: 05/15/2026)